

A Practitioner's Guide to Hyperparameter Optimization in Optuna

Report Outline

Phase 1: Algorithm Exploration and Analysis – (Completed)

The first phase of this research will explore and develop an understanding of the common algorithmic approaches to hyperparameter optimization. It will cover Bayesian Optimization, Bandit-Based, Evolutionary, and Trajectory-Based methods, which are among the most widely used strategies for black-box optimization. Black-box optimization refers to optimization problems where little to no assumptions can be made about the objective function beyond basic domain constraints. The goal of this section is to bridge the gap between theory-based algorithms and a practitioner's understanding of machine learning and statistics. Once the baseline understanding of how each algorithm style works has been developed, the focus shifts to the theory behind specific algorithms. In this study, we examine four key algorithms: the Bayesian methods, Tree Parzen Estimation and Gaussian Process, the Evolutionary method, Covariance Matrix Adaptation Evolutionary Strategy, and the Bandit-Based method, Hyperband. The goal is to build a practical understanding of the underlying logic behind each when applied to black-box problems such as hyperparameter optimization.

Phase 2: Algorithms and Algorithm Parameters in Applications – (Completed)

This phase focuses on showcasing how each algorithm is optimized and implemented in Optuna. The aim is to show how theoretical concepts translate into real-world hyperparameter optimization. Once the implementation is covered, we break down the parameters that control each algorithm, along with their typical use cases based on factors like input dimensionality (e.g., high-dimensional spaces with more than 20 parameters), search space topography (noisy and non-stationary vs. smooth and stable), and resource budgets (e.g., GPU time). This section serves as a practical reference for how each algorithm is designed to operate and why certain parameters are made available to users. We'll also note cases where specific methods, such as trajectory-based approaches, are generally avoided in resource-intensive settings like hyperparameter optimization.

Phase 3: How to Effectively Evaluate a Search Space – (In Development)

This phase aims to define a systematic approach for understanding the topography of a search space without burning through large amounts of resources. It acts as a preliminary step before full-scale hyperparameter optimization, helping guide algorithm selection with better context. The focus will be on combining domain knowledge with targeted, low-cost tests to assess key attributes such as noise levels, stationariness, the presence of local optima, and parameter interactions. These evaluations will rely on lightweight probing methods, which will be finalized based on early testing results.

Phase 4: Applying Domain and Parameter Knowledge to Real Problems – (Planned)

This phase will put our earlier insights into practice, showing how they affect solution convergence, resource usage, and overall solution quality for each Optuna sampler. The test problems will vary in model complexity, dataset size, data type (numerical, image, text, etc.), resource constraints, and domain-specific challenges such as dirty data or multi-objective goals. Before running experiments, we'll apply the search space evaluation method to map the characteristics of our search space and hyperparameter domain. Using these findings, we'll start with One-Factor-At-a-Time (OFAT) variations to measure the controlled impact of specific parameters and validate the assumptions from Phase 2. Once single-parameter effects are clear, we'll expand to studying parameter interactions to uncover more complex relationships.

Phase 5: Applications and Theoretical Evaluation – (Planned)

Using the benchmarking results, we'll evaluate how well each algorithm, and its key parameters, performs on the tasks it was designed for. This will either confirm or challenge the heuristics laid out in Phase 2. From this analysis, we'll develop guidelines for future hyperparameter optimization applications, backed by real data-driven insights from the statistical and comparative analyses conducted on the collected benchmarking results.

Deliverables & Timeline:

Deliverables:

- **Technical report** bridging the gap between theory-based algorithm design and practitioner-focused implementation in Optuna.
- **Reference tables and parameter guides** for each algorithm, outlining parameter effects, tuning heuristics, and recommended defaults.
- **Systematic selection framework** for choosing algorithms and their parameters based on search space characteristics, dimensionality, and budget constraints.
- **Benchmark results & datasets** comparing algorithm performance across varied models, data types, and resource settings.
- Provide baseline to turn algorithmic selection into an **optimization problem**

Timeline & Figures:

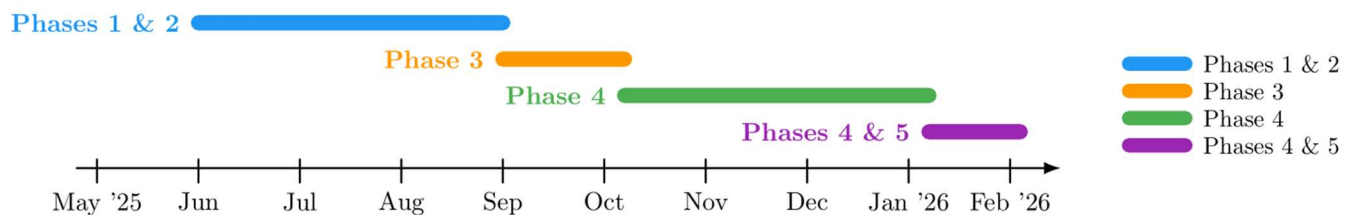


Figure 2: Research Timeline

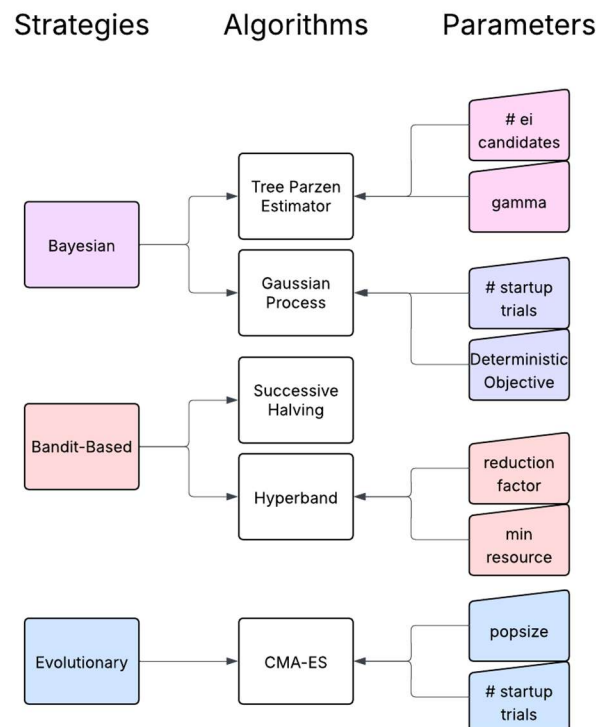


Figure 1: Phases 1 & 2 Deliverables